Encoder-first extraction with tiered, blind verification for MDS form filling

Methodology draft, 5 November 2026. Design proposed by the project owner; supporting results are from the cited studies and have not been reproduced here. Tests run with gpt-5-nano (low reasoning effort) and local encoders.

1. Components: three neural models, the LLM only for relations

Job	Done by	Role on MDS data
Mention tagging, assertion, subject, section, time spans	One clinical encoder (Clinical ModernBERT) read once, several heads	Tags every sentence of the discharge summary; each mention carries negation, who it is about, and timing. Heads learn from open labelled data (NegEx/ConText) and are applied to MIMIC unchanged.
Synonyms and item class	Bi-encoder candidates (SapBERT / BioLORD), cross-encoder rerank; LLM decides class membership	Only class membership is needed (is "Lovenox" evidence for the anticoagulant item?). The LLM reads the item definition, so no drug lists; answers are cached per normalised string.
Abbreviations	Encoder detects; candidate senses; encoder picks; low margin goes to the LLM	Expansion is the hard part. A within-record consistency feature is preferred to a hard one-sense-per-note rule.
Dates	Encoder classifies the expression and its anchor (note, admission, discharge, event); code does calendar arithmetic; LLM only for compositional cases	Window checks against the assessment reference date (ARD). Anchor choice is an explicit, measurable step; every result must fall inside the admission dates. MIMIC notes mask dates, so record timestamps carry most of the dating.
Relations (event frames)	LLM	Medication (drug, action, given / ordered / planned, indication), treatment (type, status, setting, duration), diagnosis (status, documenter, links to treatments).

2. Two levels of slot filling

- **Level 1:** fill event frames from the notes, encoders first and the LLM for the gaps. One medication frame feeds many N items, and Level 1 can be tested on open data without MDS labels.
- **Level 2:** answer each MDS item as a query over the frames and their time intervals. An active diagnosis becomes a graph computation: a documented diagnosis plus a linked medication or treatment inside the window (exact criteria from the RAI manual). The frame schema is designed by hand; the frame-to-item mapping stays model-decided.

3. Verification: yes and no are checked differently, and blind

- **Yes draft** (has a cited sentence): does the sentence say the patient received it (not held, not family)? Is the date inside the look-back before the ARD? Does the concept belong to the item class?
- **No draft** (nothing found): a wider search with a lower tagging threshold, every medication or treatment mention, negated mentions and near misses. A no draft that the graph expects to be yes (a DVT diagnosis but no anticoagulant) is sent to the wide search first.
- **Three answers:** Yes, No, Unknown. No mention counts as No only if the notes cover the window; for the 29 pilot items answered for the hospital stay rather than the nursing-home window, the honest answer is Unknown.
- **Blind check:** the LLM gets the item definition and the evidence pack, never the draft, and quotes support for Yes, for No and for Unknown (or none). Agreement is accepted; disagreement is escalated with both sets of evidence.

Tier	Who	Settles
1	Encoders and code	Clear cases: the cited sentence entails the answer, the subject is the patient, the date is clearly inside or outside the window, confidence is high.
2	Blind LLM	Unsure items, relative dates, class membership, and draft/LLM disagreements that more evidence resolves.
3	Nurse (labelling web UI)	Real conflicts, Unknown answers, anything still uncertain, shown with both evidence paths.

4. How it is tested

Layer	Data	Measure
Assertion, subject, timing heads	NegEx/ConText (has experiencer: checked); E2's held-out 360 mentions and their rewrites	Accuracy against E2's LLM and ConText arms; share settled at Tier 1
Medication spans	Glaucoma medication lines	Span F1 against E1/E13a
Abbreviations	MeDAL (open); CASI (download blocked so far)	Expansion accuracy; share sent to the LLM
Time	PMOA-TTS timelines (silver, LLM-made)	Offset error, anchor error, ordering concordance
Whole tiered system (smoke)	Subset of the 100 MIMIC pilot admissions, 29 items	Tier shares, Unknown share, blind vs shown-draft agreement, cost; table proxy labels as a weak reference (circular for record evidence)

Limits. The heads are trained (a departure from training-free); i2b2 2010/2012 and n2c2 2022 need data-use agreements and are not used; whether multi-task heads interfere is untested with a frozen encoder; gpt-5-nano at minimal reasoning misread "No evidence of pneumonia", so low effort is used.